SUPAKORN UDOMSINTUWAT

Full Stack Engineer

Chiang Mai, Thailand | +66-80-865-6423 | supakornboon5039@gmail.com
github.com/supakorn5039-boon | linkedin.com/in/supakorn-udomsintuwat

PROFESSIONAL SUMMARY

Full Stack Engineer with nearly three years of experience taking products from architecture through production. Works across **React** and **TypeScript** frontends, **Go** and **Laravel** backends, **PostgreSQL / TimescaleDB** data systems, **CI/CD**, and **cloud infrastructure**. Recent work includes the backend of a national facial detection platform at about 70 million records and 1–2 million new events per day, and a real-estate platform built and run independently on AWS.

KEY COMPETENCIES

Languages: TypeScript, JavaScript, Go, PHP, Python, SQL
Frontend: React, Next.js, Redux, Zustand, Tailwind CSS, MUI, shadcn/ui, Headless UI, React Hook Form
Backend: Go (Gin, Fiber), PHP (Laravel), Node.js (Express), Python (Flask, FastAPI), REST API design, MQTT
Data: PostgreSQL, TimescaleDB, MySQL, MongoDB, Elasticsearch, Redis
DevOps & Cloud: Docker, Kubernetes, GitHub Actions, GitLab CI, Nginx, AWS (EC2, RDS), Cloudflare, Vercel, ELK
Testing: Playwright (end-to-end), Vitest, React Testing Library, PHPUnit, Go testing, Postman (API testing)
AI / ML: Claude Code, LLM prompt workflows, image processing (OpenCV, YOLOv8), deep learning fundamentals

PROFESSIONAL EXPERIENCE

FireExit Cooperation Co., Ltd. **Oct 2024 – Present**
Computer equipment and software · Chiang Mai

Full Stack Engineer **Jan 2026 – Present**

CIB Vision – National Facial Detection System: Built the backend that processes facial detection data from nationwide CCTV feeds for Thai Police operations.

- Kept burst traffic off the request path: detection webhooks are acknowledged immediately, while deduplication, enrichment, persistence, and Telegram alerts run on **Redis-backed workers**.
- Moved detection data to **TimescaleDB hypertables** with seven-day compression and continuous aggregates as production grew to about **70 million records** and **1–2 million new events per day**.
- Cut person-search latency from **10+ seconds to under one second** by measuring index behavior on production data and replacing ineffective GIN indexes with composite and partial indexes plus aggregate-backed time-bucket search.
- Removed **foreign-key bottlenecks** from the compressed detection table and added guarded pruning of stale master data; the production cleanup finished in seven seconds without touching detection records.
- Enforced server-side, camera-level access for about **1,800 cameras across 3,700 police units**, combining role permissions, organization grants, manual overrides, and ThaiID identity data.
- Ran the system on CIB infrastructure with **Docker, Kubernetes** workers and scheduled jobs, **GitLab CI**, Harbor, SonarQube, and ELK logging that excludes PII.
- Maintained backend and end-to-end regression suites with **Playwright**, mocking integrations that are only reachable through the government VPN.

Bangkok Patana School – Heat Index Monitoring System: Built the Laravel backend and React frontend that turn LoRaWAN sensor data into live heat-risk readings, and had it in production within four weeks of kick-off.

- Replaced the planned InfluxDB and Laravel Reverb with a **TimescaleDB** hypertable, continuous aggregates, and a small **Go WebSocket relay** fed by Redis, cutting the stack from ten services to eight.
- Received sensor data over **MQTT** with TLS and per-topic ACLs, plus an HTTPS push route for the gateway behind the school's SIM router, rejecting readings outside the sensor's physical range.
- Rewrote Excel/CSV export as a stream after a year of data exhausted PHP's memory limit; **1.05 million rows** now export in **29 seconds using 32 MB**.
- Built an **OAuth2 client-credentials API** with Laravel Passport so the school's own systems can pull hourly heat-index averages, alongside role-based access control and an audit log.

- Hardened sign-in with per-IP and per-username rate limits and session revocation on password change; **237 PHPUnit tests** run in **GitLab CI** before every QA deploy, plus a Playwright end-to-end suite.

Frontend Developer **Oct 2024 – Jan 2026**

- Designed an internal **UI component library** (shadcn/ui, Headless UI, MUI, Tailwind CSS), including complex data-driven elements such as advanced data tables, and led its adoption across multiple project teams, **cutting feature delivery cycles by 30%**.
- Developed **real-time IoT monitoring interfaces** that combine device telemetry and network status from backend APIs.
- Built interactive dashboards for **RMUTI Central Lab** to track water and electricity consumption with dynamic charts.
- Implemented **vehicle-tracking dashboards** for Pandora Logistics with real-time entry/exit monitoring and synchronized updates.
- Delivered responsive interfaces for the **Mochit Room Booking System**, covering scheduling logic, user availability, and API integration.

Wealthy Prime Estate — May 2026 – Present

Freelance · wealthyprimeestate.com · client production handover Aug 2026

Full Stack Engineer (Freelance)

Independently took a property-brokerage platform from requirements and architecture to production handover in about 10 weeks, and continue to operate it for the client.

- Chose **Go (Gin)** and **PostgreSQL** for an API covering property search, bookings, agent workflows, administration, financial exports, and audit history.
- Delivered the **React 19** frontend in strict **TypeScript** with separate customer, agent, and admin flows and full Thai, English, and Chinese localization.
- Kept owner contact details and documents out of every public response; only assigned agents and administrators can read them.
- Used **HMAC-signed** property and agent identifiers for tamper-resistant review links, so customers can leave verified post-visit reviews without an account.
- Moved the backend and database from US-West to **AWS Singapore**, cutting API latency for Thai users by about **150–200 ms** while preserving existing JWT sessions.
- Run production for about **$30 per month** on ARM EC2, private encrypted **RDS PostgreSQL**, Cloudflare, and R2, with nightly off-site backups and network-isolated database access.
- Set up **GitHub Actions** deployment with just-in-time SSH rules that authorize only the runner IP and are revoked after every deployment, including failed runs.

Ocare Health Hub Co., Ltd. — Jan 2024 – Sep 2024

Software platform · Chiang Mai

Software Engineer (Frontend)

- Developed a patient-facing web platform in **React** and **Next.js** that gives patients instant digital access to medical reports, reducing waiting times for results.
- Built an **automated clinical report delivery workflow** that reduced administrative work for medical staff.
- Built **dashboards** for medical professionals to review patient history and diagnostic results, synchronized in real time across patient and doctor interfaces through secure REST APIs.
- Improved **LLM prompt workflows** that turn natural-language questions into MongoDB queries for retrieving patient records.

EDUCATION & LANGUAGES

Bachelor of Engineering (B.Eng.)
Information Systems and Network Engineering (International Program)
Chiang Mai University · 2020 – 2024

English
CMU E-Graduate: **78/100** (Upper-Intermediate, B2)

CERTIFICATIONS & TRAINING

Cybersecurity Essentials
Cisco Networking Academy · Mar 2024

CCSK v4.1 Foundation Training
Cloud Security Alliance · Feb 2023

Full-Stack Web Development
LinkedIn Learning · Dec 2023
Getting Started as a Full-Stack Web Developer; Learning Full-Stack JavaScript Development: MongoDB, Node, and React